

WEATHERFORD STAFFORD, OK, USA

WMO: 722368

Lat: 35.550N

Lon: 98.667W

Elev: 489

StdP: 95.58

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 00364

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-7.6	-17.0	0.9	-9.1	-13.9	1.2	-5.9	12.6	5.0	11.7	5.6	4.1	0	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	37.9	22.9	37.1	22.9	35.3	22.9	25.0	32.8	24.4	32.3	23.9	31.8	5.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.8	18.6	27.4	22.4	18.2	26.9	22.2	17.9	26.7	78.7	32.8	76.3	32.9	74.4	32.0	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.5	10.4	9.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	3.3	5.2	10.3	14.8	20.2	26.5	28.1	27.2	23.1	15.7	9.2	4.0
	DBStd	10.12	5.51	6.43	5.83	4.97	4.72	3.05	3.23	3.39	3.94	5.37	5.07	4.92
	HDD10.0	734	214	159	66	15	1	0	0	0	0	14	72	194
	HDD18.3	2093	466	369	253	126	36	0	1	0	7	115	275	445
	CDD10.0	2811	7	24	76	158	318	497	561	533	393	190	48	7
	CDD18.3	1128	0	1	5	19	95	247	302	274	151	32	1	0
	CDH23.3	13017	1	13	59	226	958	2892	3813	3216	1512	312	16	0
	CDH26.7	6506	0	2	12	53	374	1463	2129	1730	661	81	2	0
Wind	WSAvg	4.1	4.0	4.3	4.8	5.1	4.5	4.7	3.6	3.4	3.5	4.0	4.1	3.7
Precipitation	PrecAvg	692	23	28	51	54	108	97	51	76	78	73	41	27
	PrecMax	1072	78	106	172	219	418	285	160	188	234	231	175	95
	PrecMin	375	0	0	0	5	8	10	0	8	1	3	0	0
	PrecStd	173	23	25	40	40	83	58	40	48	60	66	41	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.1	25.1	28.0	31.1	35.0	38.9	41.0	40.8	37.2	31.4	26.0	19.9
		MCWB	10.8	13.4	16.3	17.8	21.4	22.8	23.0	23.1	21.9	19.9	16.0	11.3
	2%	DB	17.6	21.3	25.0	27.8	32.5	37.0	37.9	37.9	34.8	28.9	22.2	17.2
		MCWB	8.9	12.7	15.0	17.2	21.0	22.6	23.0	23.2	22.5	19.7	14.5	11.4
	5%	DB	15.1	17.8	22.3	26.1	31.1	35.2	37.1	37.1	32.4	27.1	19.9	14.7
		MCWB	8.1	11.4	14.2	16.6	20.6	22.9	23.0	23.2	21.7	18.4	13.6	8.9
	10%	DB	12.3	15.8	20.0	23.6	28.0	33.0	35.2	34.9	31.0	24.9	17.4	12.2
		MCWB	6.6	10.2	12.9	15.9	19.8	22.4	23.1	22.9	21.4	17.2	12.4	7.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	16.2	18.7	20.9	24.0	25.7	26.2	25.4	24.6	23.0	18.9	16.1
		MCDB	18.2	18.3	25.4	25.9	30.6	33.3	32.7	33.4	31.7	27.5	21.7	17.3
	2%	WB	10.1	14.4	17.3	19.7	23.2	24.7	25.1	24.6	23.8	22.0	16.3	12.4
		MCDB	16.3	19.3	22.0	25.1	29.0	32.4	32.9	33.0	30.2	26.1	18.1	14.7
	5%	WB	8.6	12.7	15.6	18.6	22.3	24.0	24.4	24.0	23.2	20.6	14.8	9.9
		MCDB	14.5	17.1	19.6	23.8	27.9	31.6	32.4	32.7	28.9	24.3	18.4	13.7
	10%	WB	7.1	10.4	14.0	17.4	21.3	23.5	23.9	23.6	22.4	19.1	13.0	7.9
		MCDB	11.7	16.0	19.1	21.5	26.4	31.1	32.0	31.8	28.0	23.3	16.7	11.4
Mean Daily Temperature Range	5% DB	MDBR	11.9	11.7	12.8	12.7	12.4	12.1	12.1	12.1	11.9	12.4	11.9	10.5
		MCDBR	16.6	16.2	15.4	16.1	14.5	13.6	13.9	14.3	13.8	14.6	14.6	14.7
		MCWBR	9.8	9.4	8.7	8.2	6.1	4.3	3.7	4.0	4.7	7.3	8.2	8.8
	5% WB	MCDBR	15.7	14.3	13.0	13.1	12.4	11.9	12.0	12.5	10.9	11.2	11.9	12.5
		MCWBR	9.7	9.5	8.5	7.5	6.2	4.7	4.2	4.2	4.3	6.4	8.4	8.9
Clear-Sky Solar Irradiance	taub	0.282	0.291	0.332	0.368	0.395	0.416	0.414	0.415	0.377	0.347	0.306	0.283	
	taud	2.523	2.508	2.409	2.337	2.334	2.317	2.343	2.292	2.374	2.443	2.496	2.518	
	Ebn at Noon	931	959	939	915	890	868	866	860	879	880	891	906	
	Edh at Noon	80	91	110	125	127	129	125	129	113	96	82	76	
All-Sky Solar Radiation	RadAvg	2.90	3.53	4.66	5.76	6.26	6.98	6.97	6.30	5.27	3.98	3.04	2.45	
	RadStd	0.28	0.49	0.32	0.44	0.64	0.58	0.39	0.38	0.37	0.52	0.40	0.25	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page